Tide retrospective: *the susceptibility is the connected variance*

Claude Opus 4.8 (1M ctx), with Daniel Murfet

2026-05-30

Abstract

The second connected cumulant (κ_2) of the perturbation observable $u(x) = -(tx)$ for the anharmonic Gibbs measure, realised as a fluctuation–dissipation identity: the derivative at $h = 0$ of the perturbed mean $\langle u \rangle_h = G_1(h)/G_0(h)$ equals the connected variance

$$\frac{\partial}{\partial h}\Big|_{h=0} \langle u \rangle_h = \langle u^2 \rangle_0 - \langle u \rangle_0^2.$$

A pointwise `HasDerivAt.div` on the arc's partition derivatives, plus a denominator-clearing `ring` identity tying the quotient form to the moments. New file `AnharmonicSecondCumulant.lean`; builds clean, no `sorry`/`axiom`/`native_decide`.

1 Setting

This is the sixth tide of a single-day anharmonic arc on the laplace seabed, and the second rung of the cumulant ladder opened by the cumulant generating function's first derivative[1]. The perturbed mean of $u = -(tx)$ is $\langle u \rangle_h = G_1(h)/G_0(h)$, where $G_n(h) = \int (-(tx))^n e^{-t(L+hx)}$. Its h-derivative at 0 — the linear response, or susceptibility — is, by the fluctuation–dissipation theorem, the connected variance of u under the unperturbed measure. It was the lowest-risk next rung: unlike the higher cumulants it needs no general-h log machinery, only a *pointwise* quotient rule at $h = 0$ on derivatives the arc already provides.

2 The thing formalised

For the anharmonic potential, $\lambda, \gamma > 0$, $\alpha^2 < 3\lambda\gamma$, $t > 0$:

$$\mathrm{HasDerivAt}\ \left(h \mapsto \frac{G_1(h)}{G_0(h)}\right)\ \left(\frac{G_2(0)G_0(0) - G_1(0)^2}{G_0(0)^2}\right)\ 0,$$

*and that derivative equals $\langle u^2 \rangle_0 - \langle u \rangle_0^2$ (the **`gibbsExp`** of $(-(tx))^2$ minus the square of the **`gibbsExp`** of $(-(tx))$).*

Writing `G n h` for `weightedPartition lam alpha gamma t n h`:

```
theorem anharmonic_mean_hasDerivAt ... :
    HasDerivAt (fun h => G 1 h / G 0 h)
      ((G 2 0 * G 0 0 - G 1 0 * G 1 0) / G 0 0 ^ 2) 0
```

[1] `2026-05-30-tide-anharmonic-logpartition-deriv`.

3 Proof strategy

The susceptibility is one application of `HasDerivAt.div` to the pointwise derivatives of G_1 (with value $G_2(0)$) and of G_0 (with value $G_1(0)$) at $h = 0$ — both instances of the arc's `weightedPartition_hasDerivAt` — using $G_0(0) = Z(0) \neq 0$. The connected-variance form then rewrites each `gibbsExp` moment as $G_n(0)/G_0(0)$ (moment-normalisation + the `iteratedDeriv` identity) and verifies the scalar identity $(ac - b^2)/c^2 = a/c - (b/c)^2$.

4 Roadblocks and resolutions

Two small ones. (i) The moment-normalisation helper takes the base point h as an *implicit* argument under a $\forall$; supplying the $|h| < 1$ proof with `by norm_num` left h as a metavariable, so `norm_num` faced $|?h| < 1$ and stalled. Pinning `(h := 0)` fixed it. (ii) For the scalar identity I first wrote `field_simp; ring`, but `field_simp` closed the goal outright on this Mathlib build, leaving `ring` with no goals. Rather than depend on that (`field_simp`'s closing behaviour drifts between cache versions), I cleared the $G_0(0)$ denominators explicitly with `div_sub_div`/`div_eq_div_iff` so `ring` always has exactly the polynomial identity to finish.

5 What was Mathlib, what was new

Mathlib. `HasDerivAt.div` and `.deriv`; the denominator-clearing division lemmas[2]; and `ring`. **Seabed (reused).** `weightedPartition_hasDerivAt`, `weightedPartition_zero_zero_pos`, moment-normalisation and the `iteratedDeriv` identity — all from this session's arc. **New.** The susceptibility theorem and its connected-variance form. About 75 lines.

6 Lessons

- **The susceptibility needs only a pointwise quotient rule.** Reading κ_2 as $\partial_h \langle u \rangle_h$ rather than $\partial_h^2 \log Z$ avoids the general-h log machinery entirely: `HasDerivAt.div` at the single point 0 suffices.
- **Pin implicit base points before discharging their side conditions.** `by norm_num` cannot prove $|?h| < 1$ for a metavariable; `(h := 0)` first.
- **Don't lean on `field_simp` to *close* a goal.** Its closing behaviour drifts between Mathlib caches; clear denominators explicitly and let `ring` finish, so the proof survives a cache bump.

7 Follow-ups

- **Third and fourth cumulants κ_3, κ_4.** These do need the general-h derivatives (differentiating the variance/mean functions on a neighbourhood), building toward the flow-equation identity (survey v4 #2).
- **Promote `amgm_t_abs_x` to `lean-common`.**

[2] `div_pow`, `div_sub_div`, `div_eq_div_iff`, `pow_ne_zero`, `mul_ne_zero`.